

- **Lab Title:** Virtual Lab Environment Setup, Connectivity Verification & Web Proxy Configuration
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- **Student ID:** 2025/FWSD/11509
- **Course Name:** BVWS102 – OWASP Top 10 Vulnerabilities & Exploitation Techniques
- **Instructor Name:** Aminu Ibrahim Jalo
- **Date of Submission:** 21 DEC, 2025
- **Version:** Version 1.1.0

LAB OVERVIEW/OBJECTIVE

1. Objective

The objective of this lab is to assess the student's ability to correctly set up a basic offensive security lab environment, verify network connectivity between attacker and target machines, deploy vulnerable web applications, and configure Burp Suite Professional with browser proxy interception

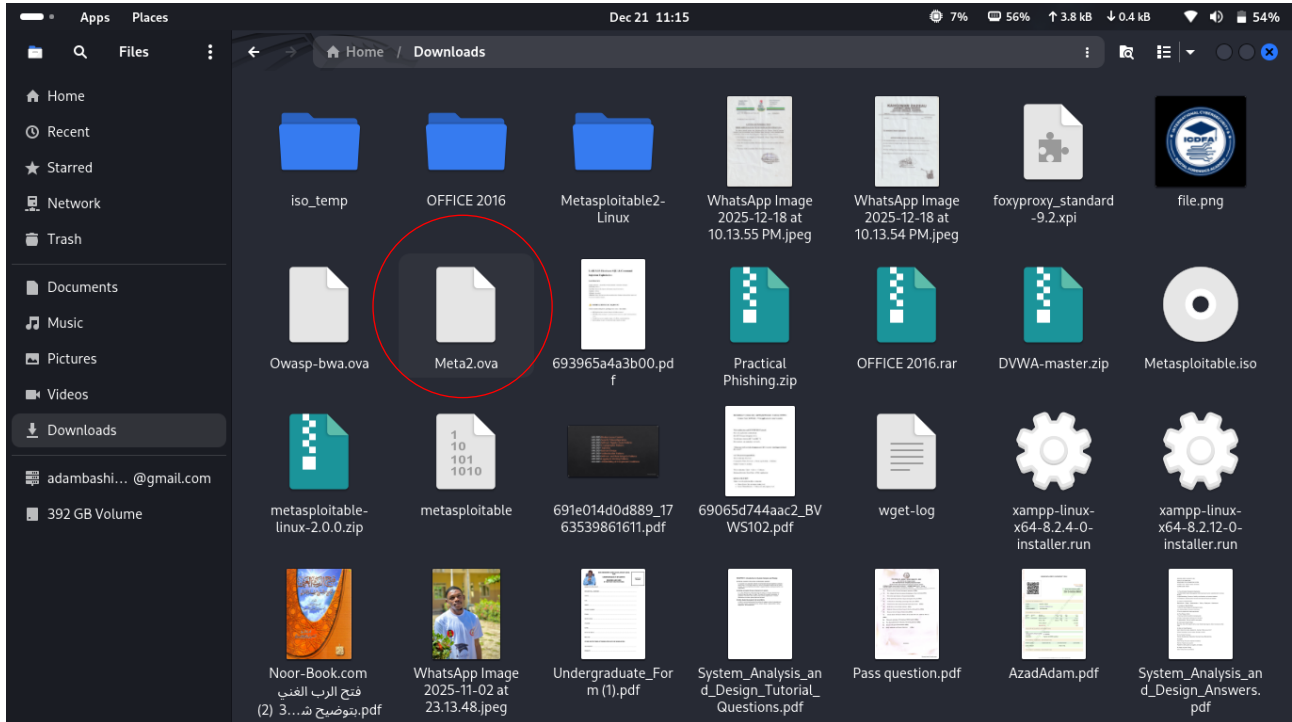
2. Lab Environment Requirements

- Each student must use the following setup:
- Hypervisor: VMware Workstation or VirtualBox
- Attacker Machine: Kali Linux
- Target Machine: Metasploitable2
- Browser: Firefox (on Kali Linux)
- Proxy Tool: Burp Suite Professional
- Vulnerable Web App: DVWA or Mutillidae

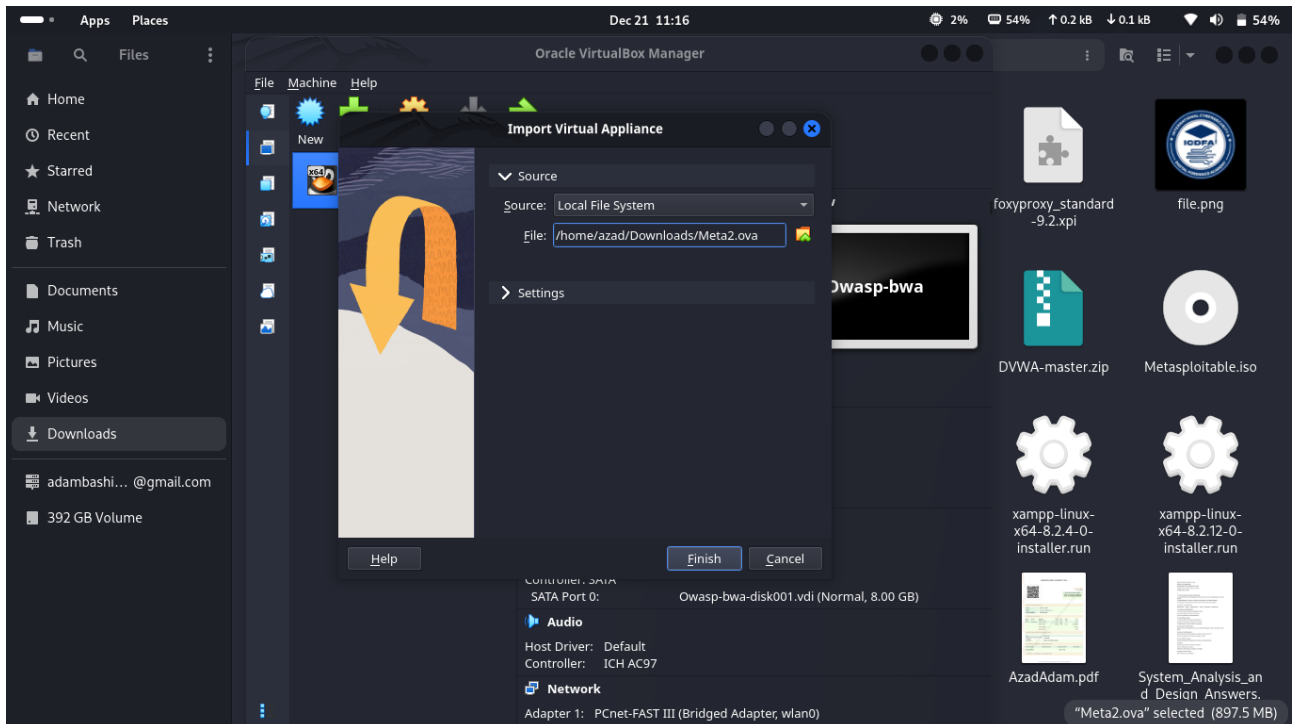
Lab Tasks & Evidence Requirements

Task 1: Metasploitable Installation

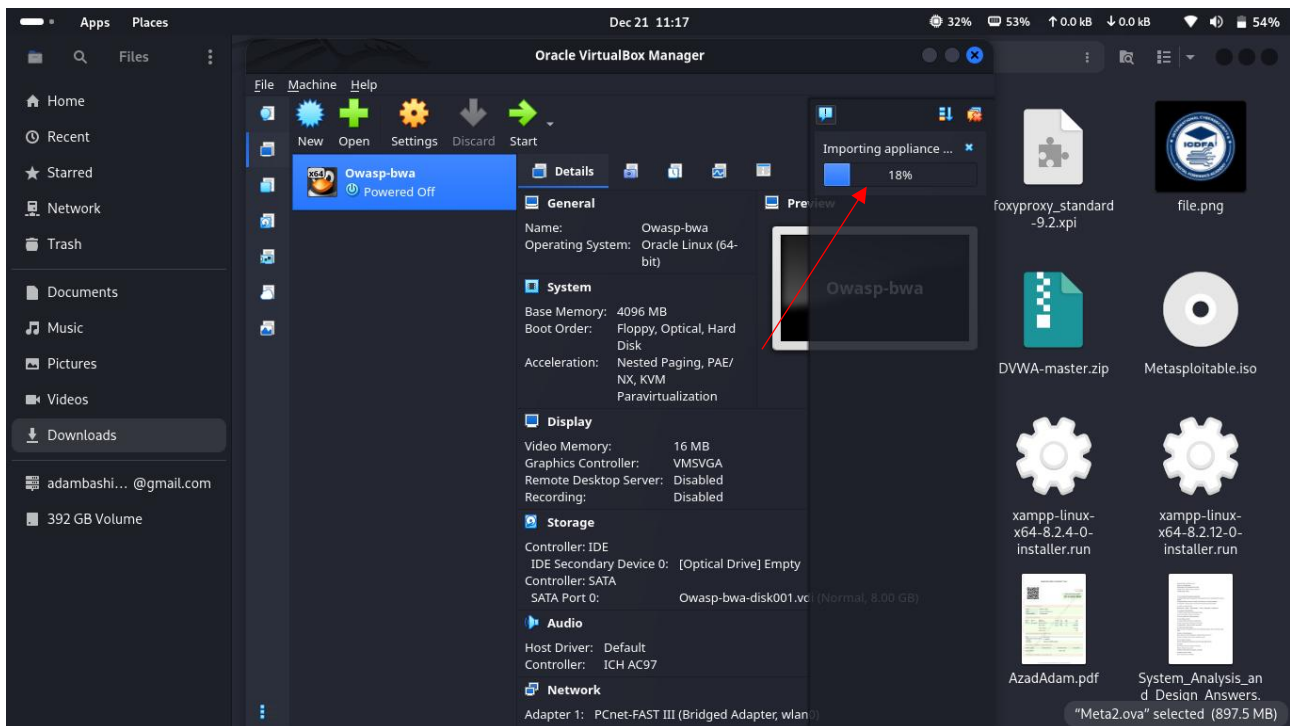
After Successful installation of VirtualBox and download of the metasploitable2 from the ICDDFA platform



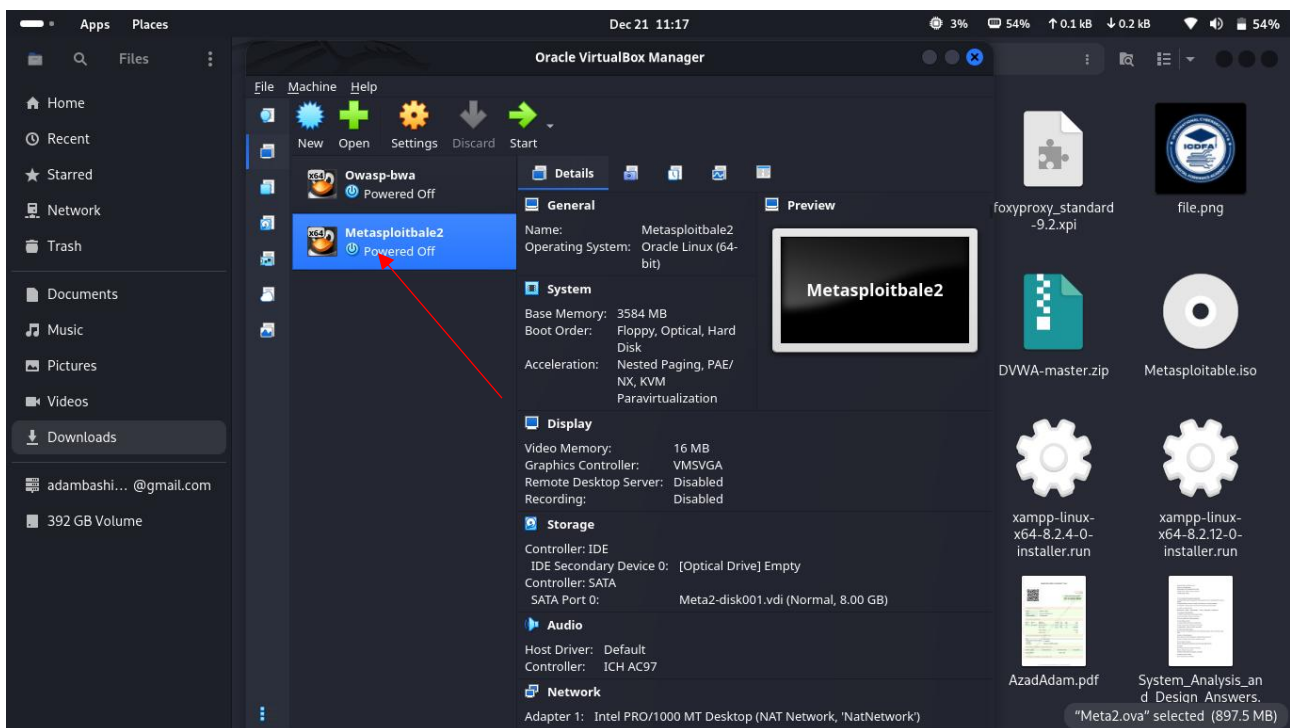
I move direct to import the file in the VirtualBox



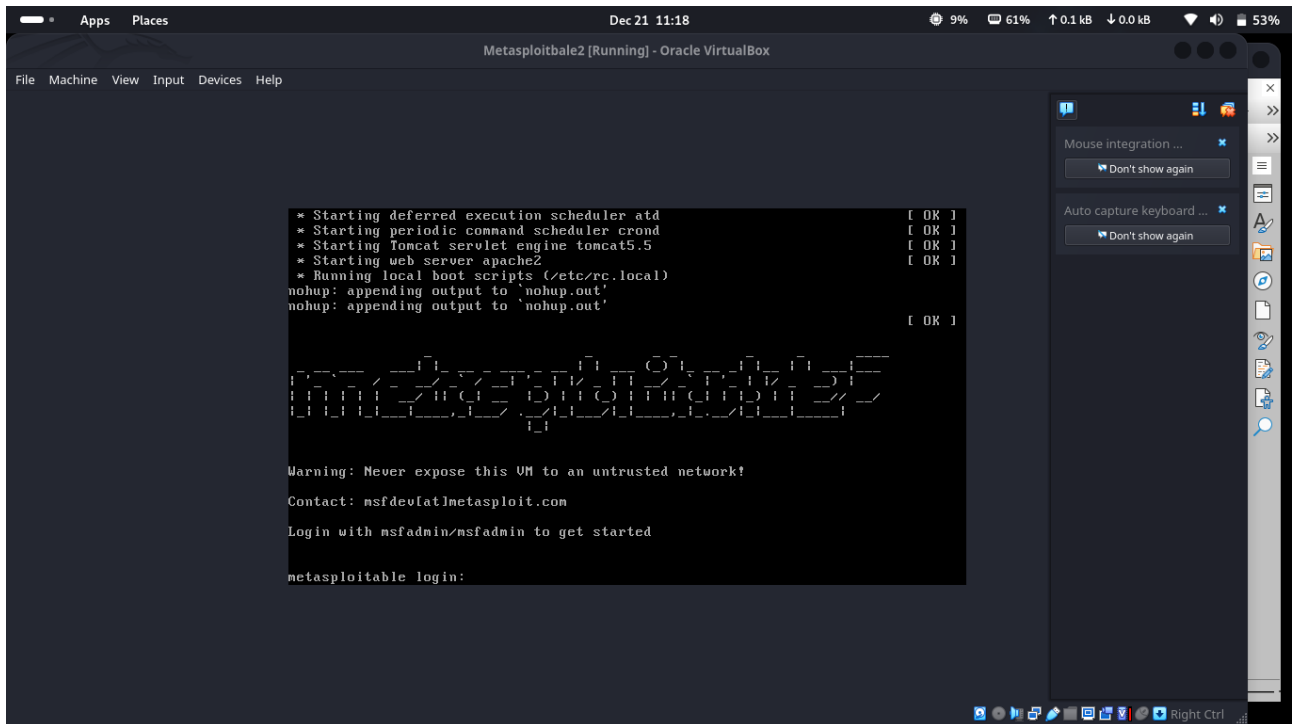
The process of importing



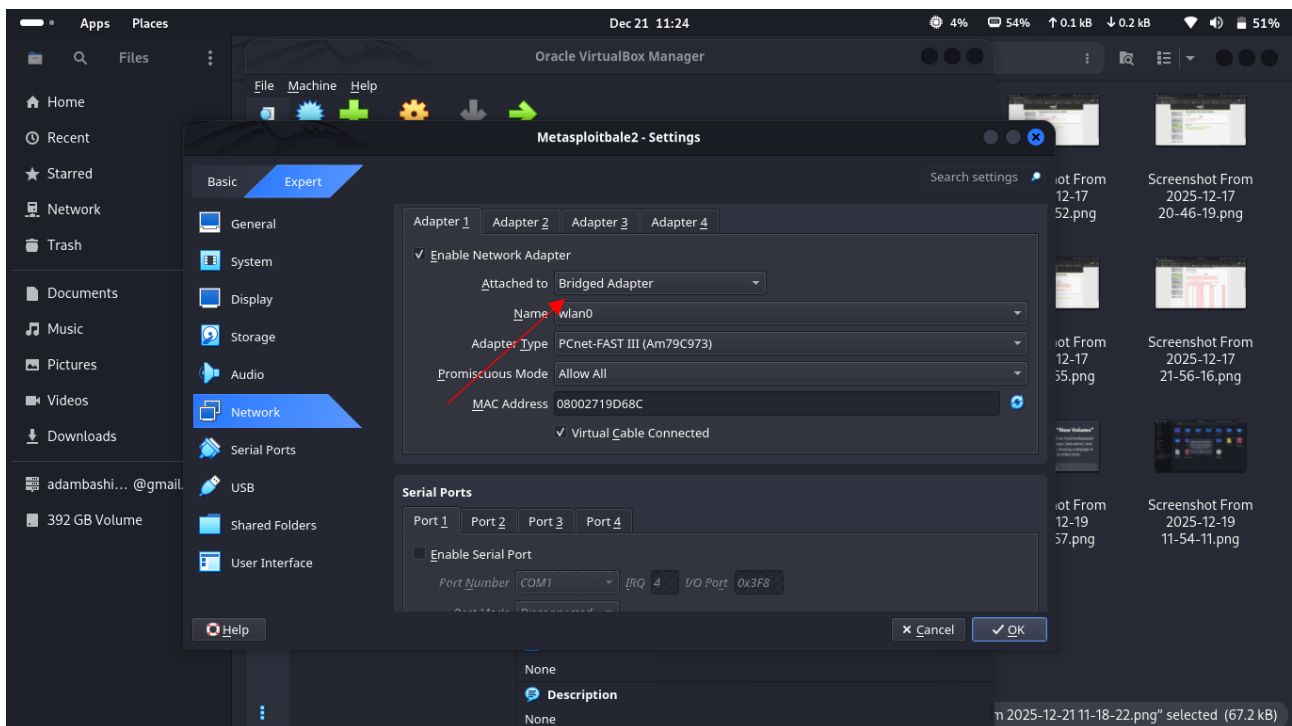
Importing Successfull



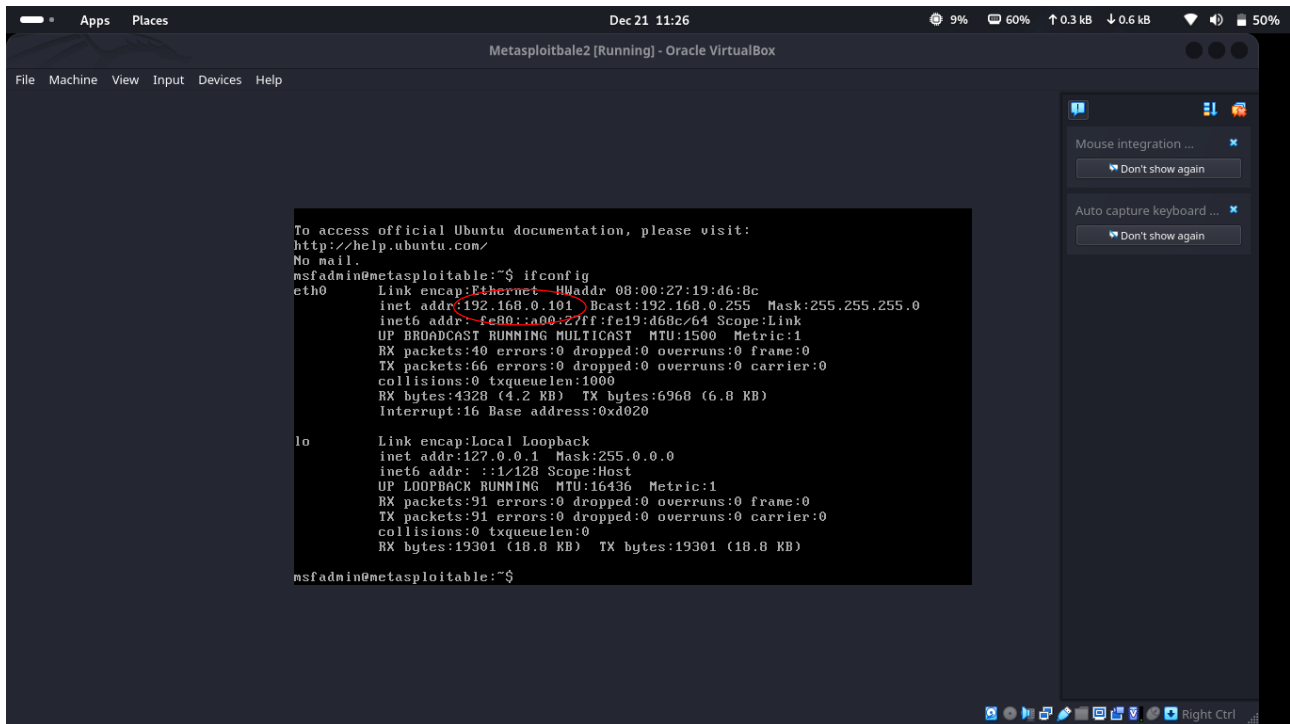
I run a quick test on the metasploitable2 and it works Successfull



I chance the network setting to Bridged Adapter because the defaults setting does not support Host Only Adapter due to the reason that am using Kali Linux as my main operating system

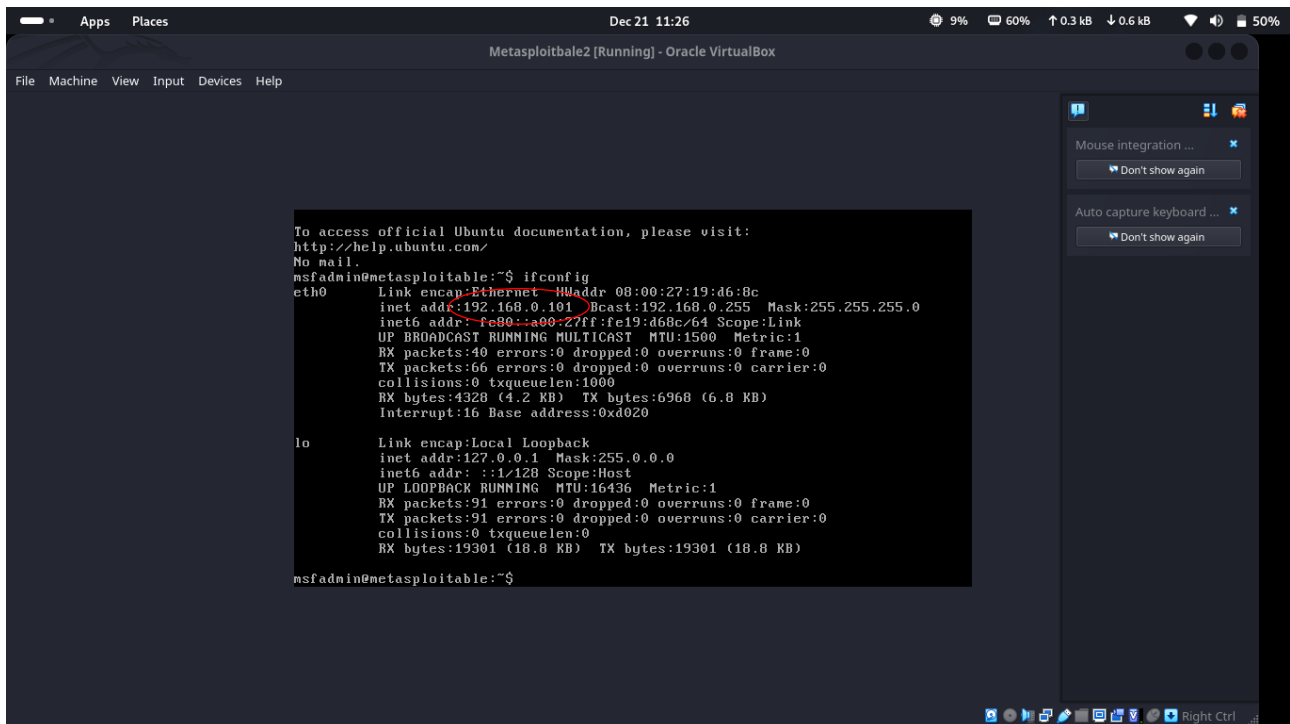


The metasploitable2 is set Successfully and the IP is ok

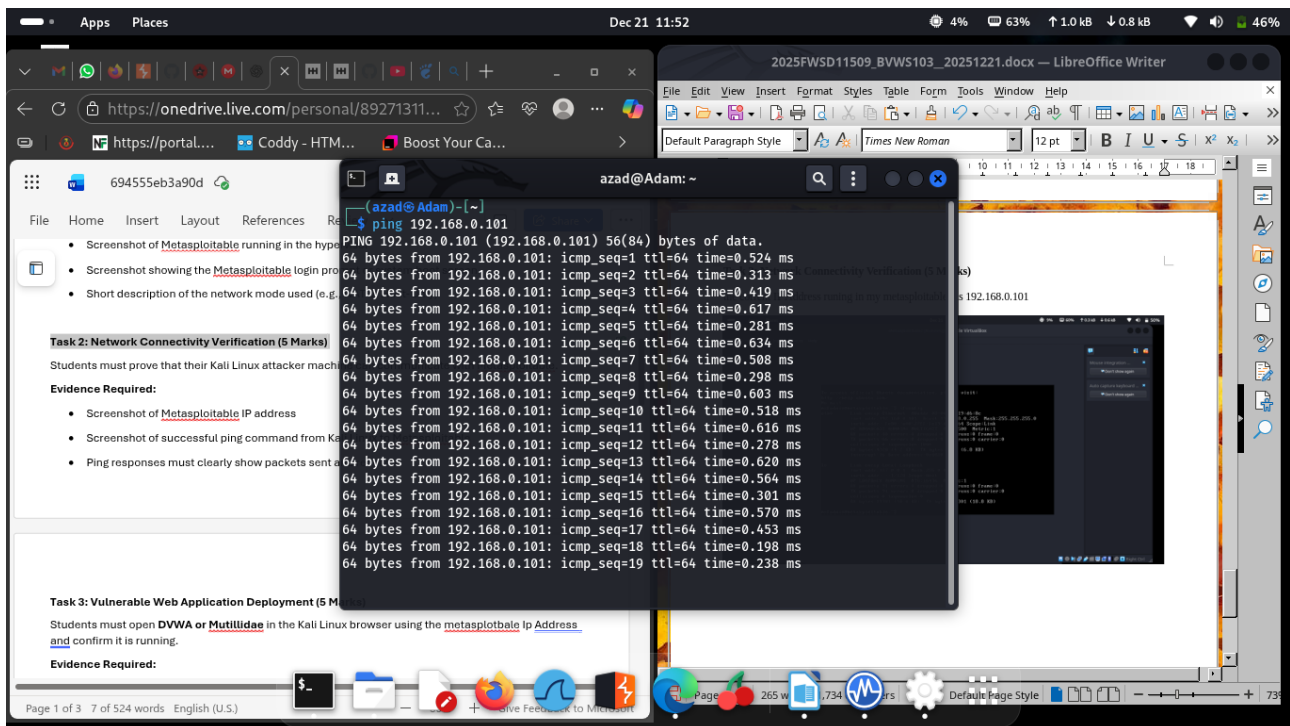


Task 2: Network Connectivity Verification

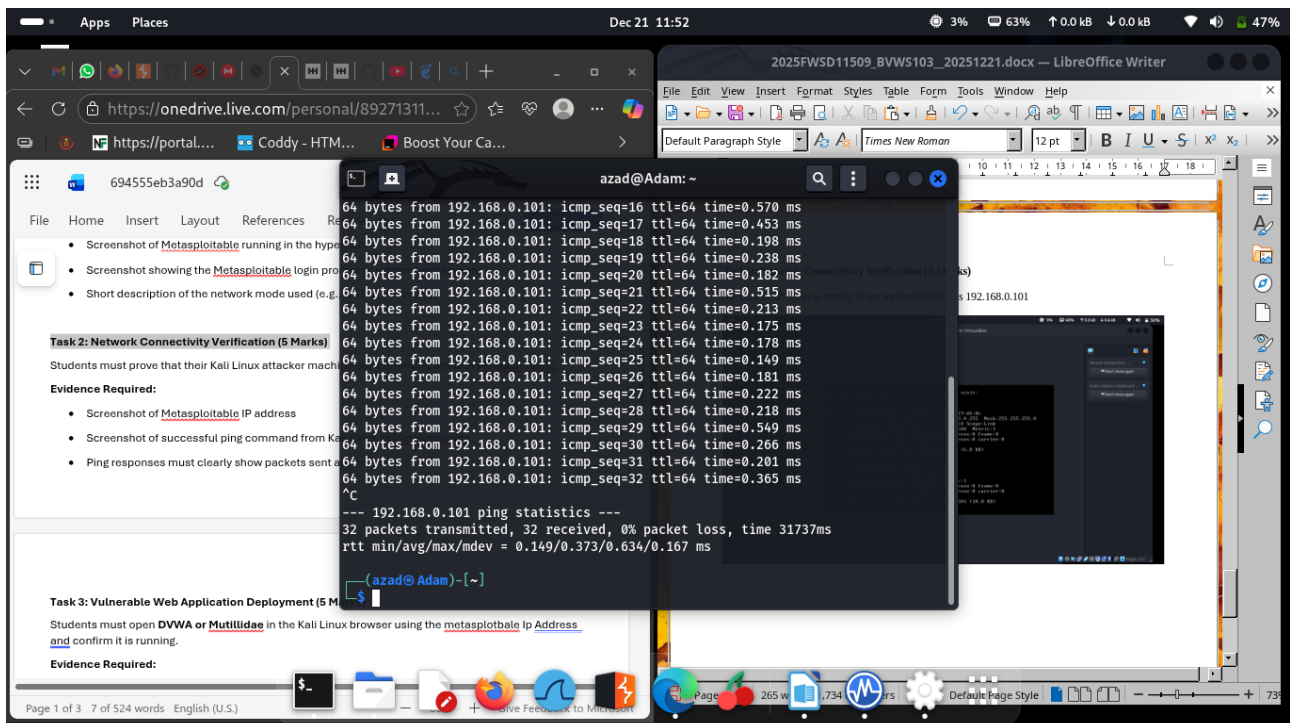
The current IP Address running in my metasploitable2 is 192.168.0.101



From my terminal I ping the IP Address using the command ping 192.168.0.101

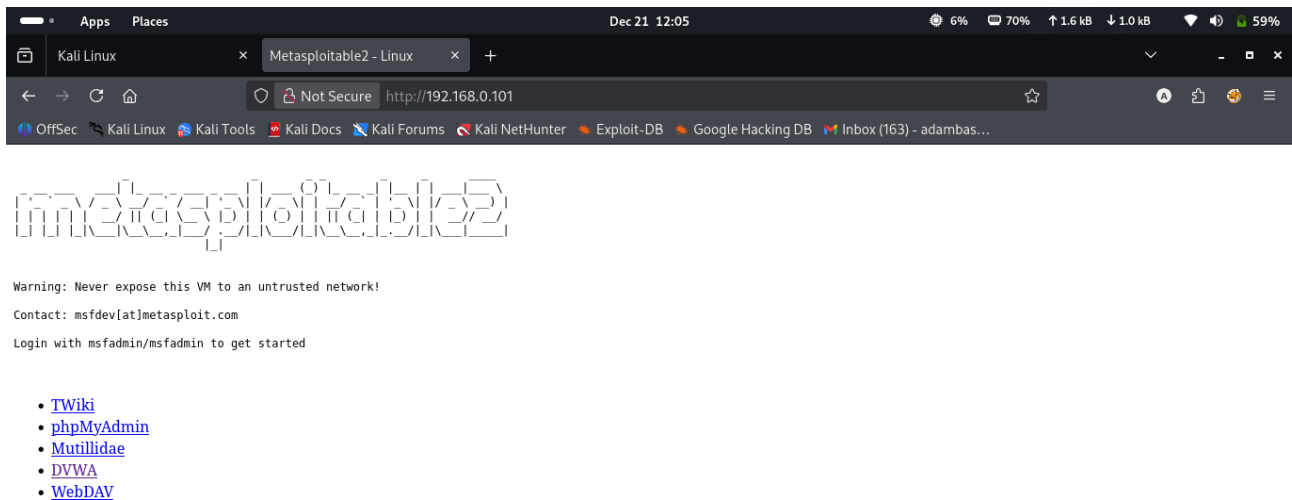


A Successfully Pinging responses showing packets sent and received

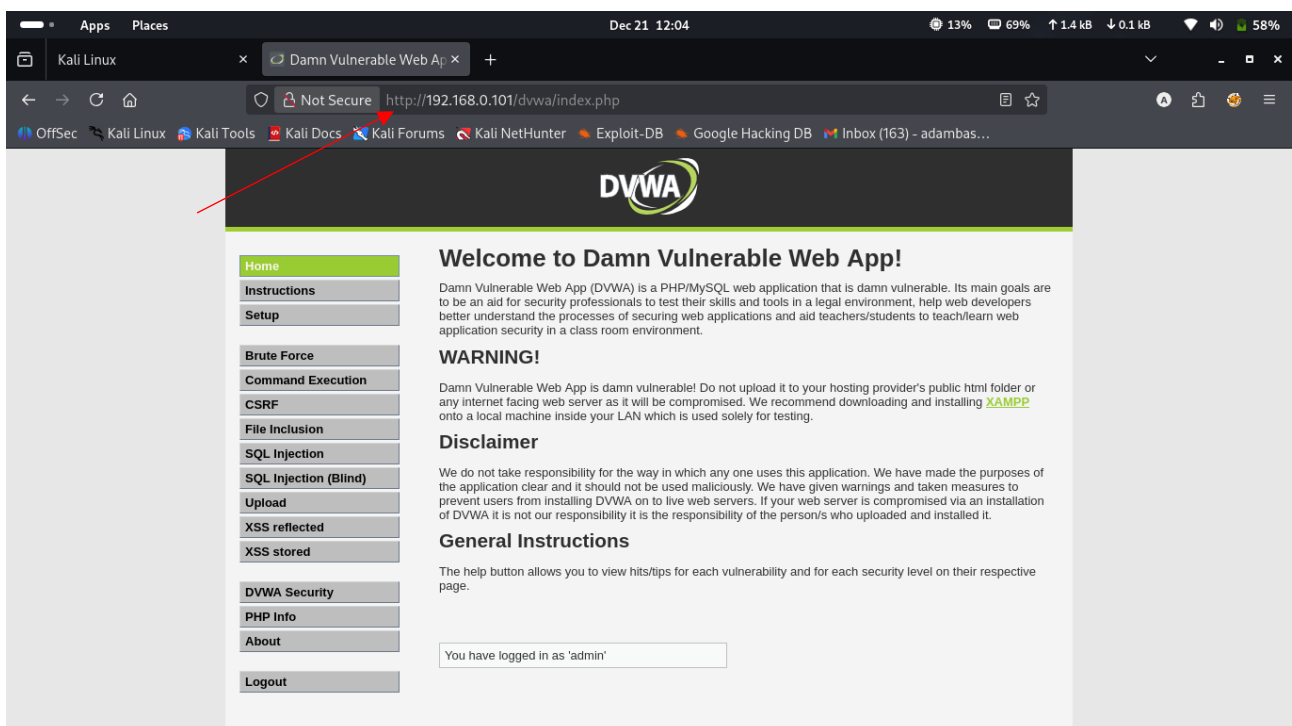


Task 3: Vulnerable Web Application Deployment

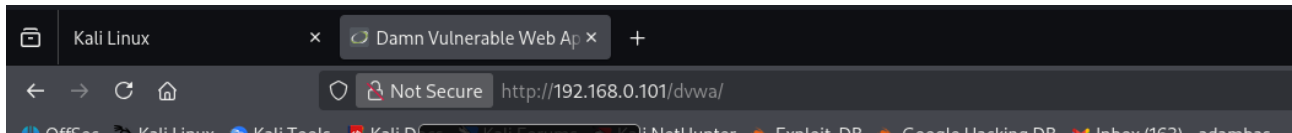
This is home page of my metasploitable2 and the options of vulnerable website it contain



DVWA is a deliberately insecure web application written in PHP & MySQL/MariaDB, used for learning and practicing web application security. It helps students and security beginners in cybersecurity to practice common web vulnerabilities legally in a safe lab environment. And it helps to Learn how attacks work, Understand why insecure code is dangerous, Practice secure coding and Prepare for CTFs, exams, and real pentesting

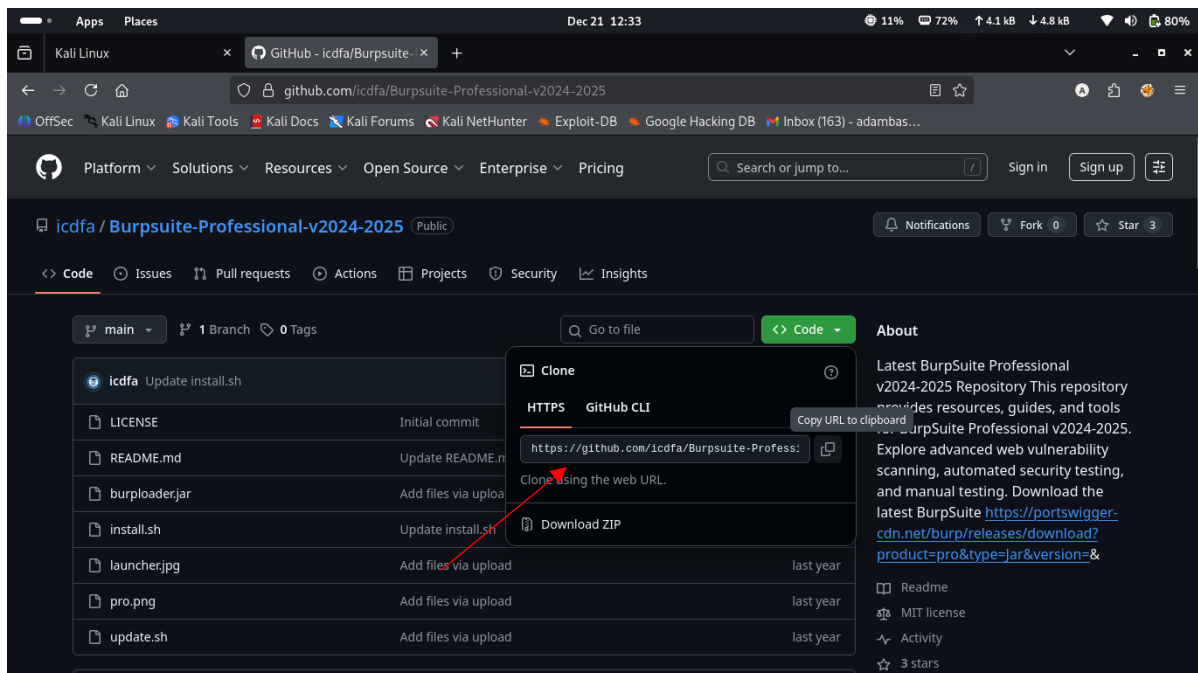


The URL bar showing the application address

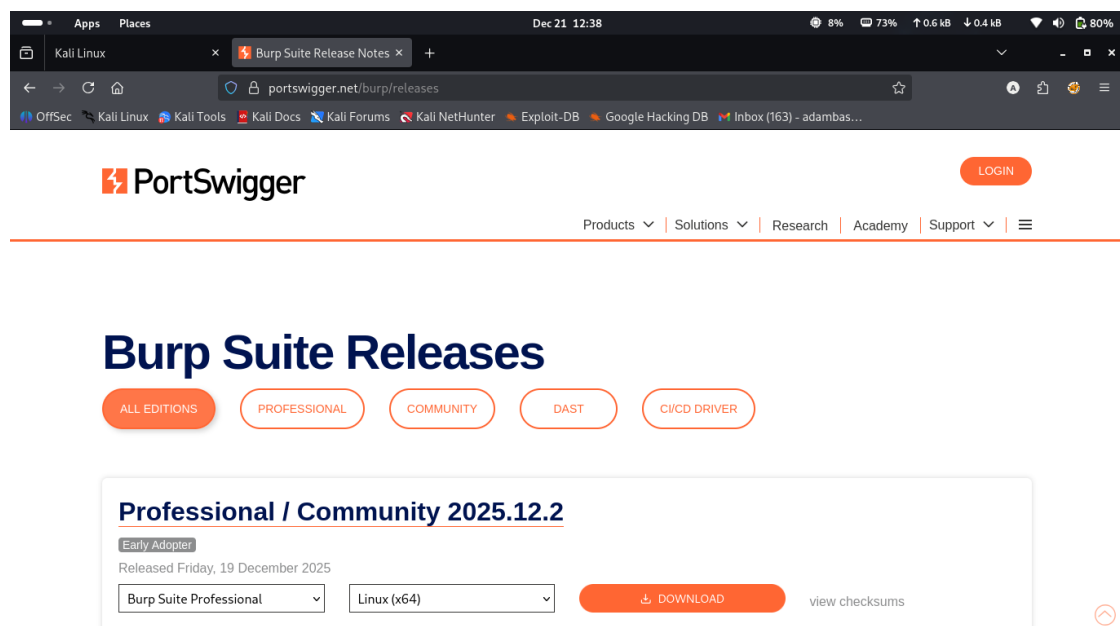


Task 4: Burp Suite Professional Installation & Licensing (7 Marks)

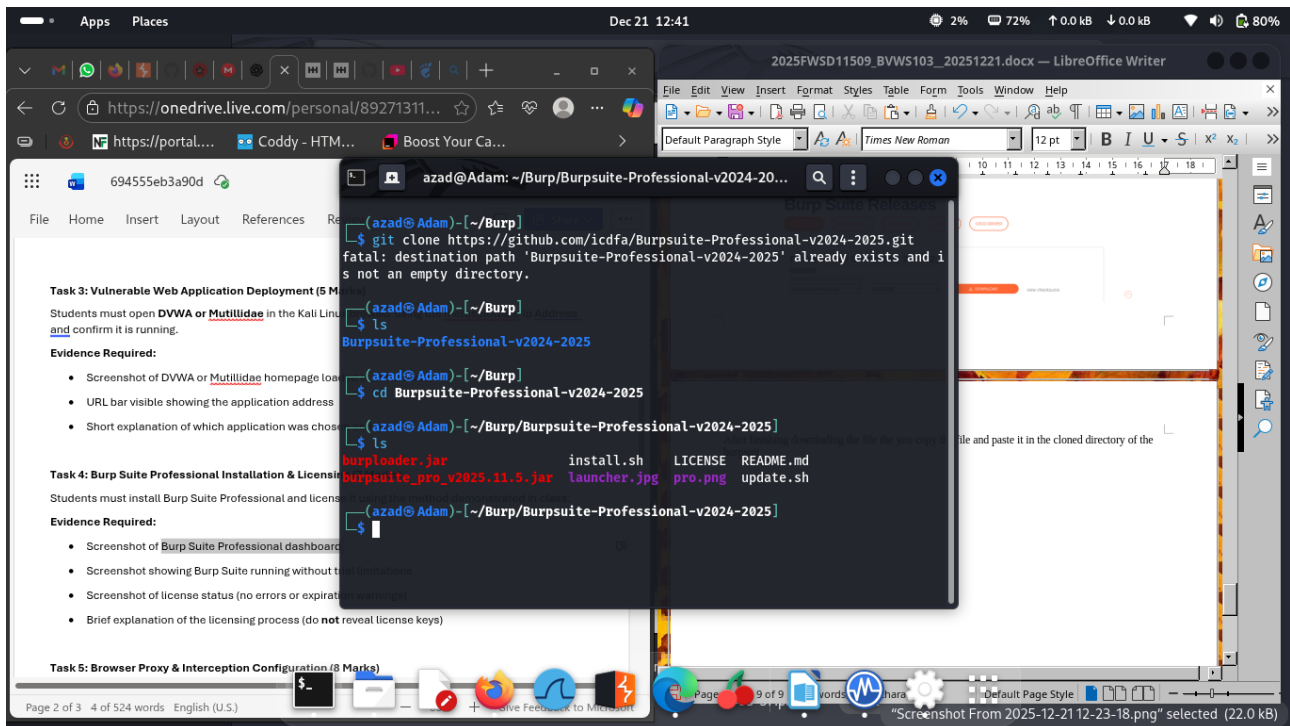
I first of all go to the ICDFA/Burpsuite-professional site and click on the code to copy the URL



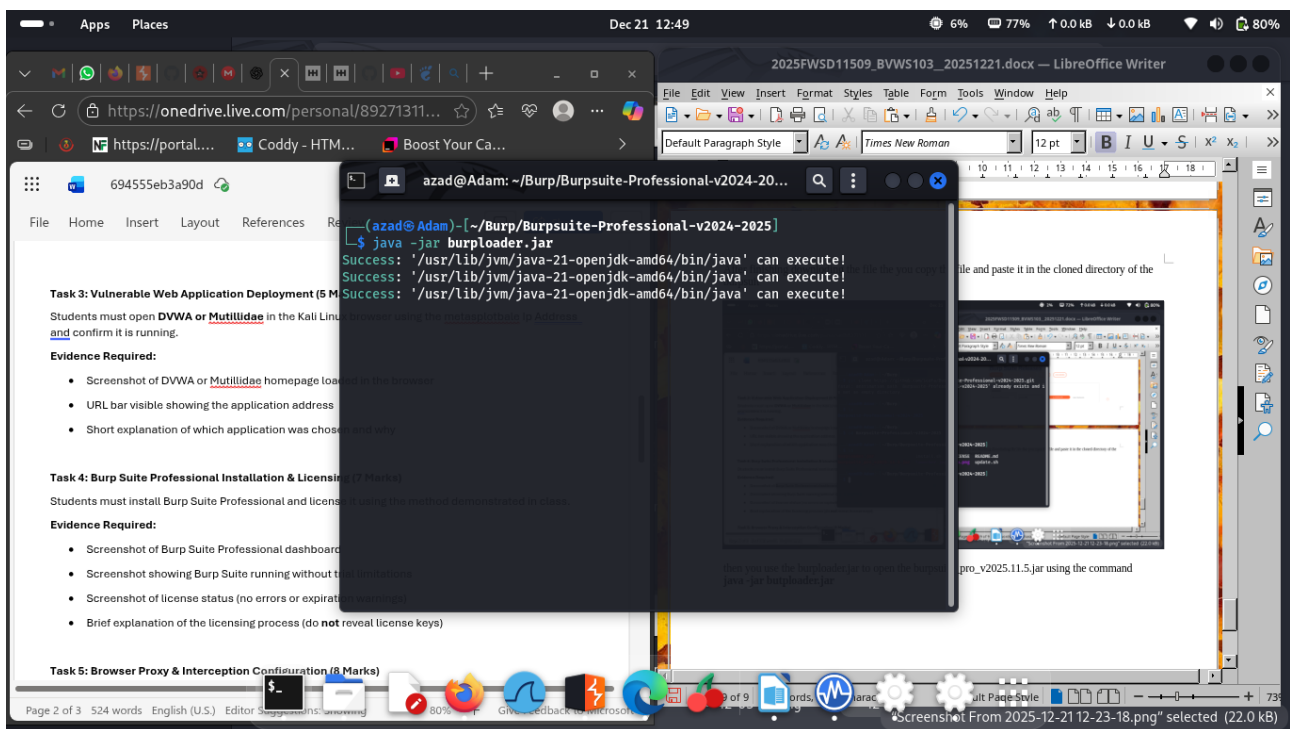
Then move to our terminal to clone it using git and the go to the Burp Suite Professional dashboard to download the java file of it



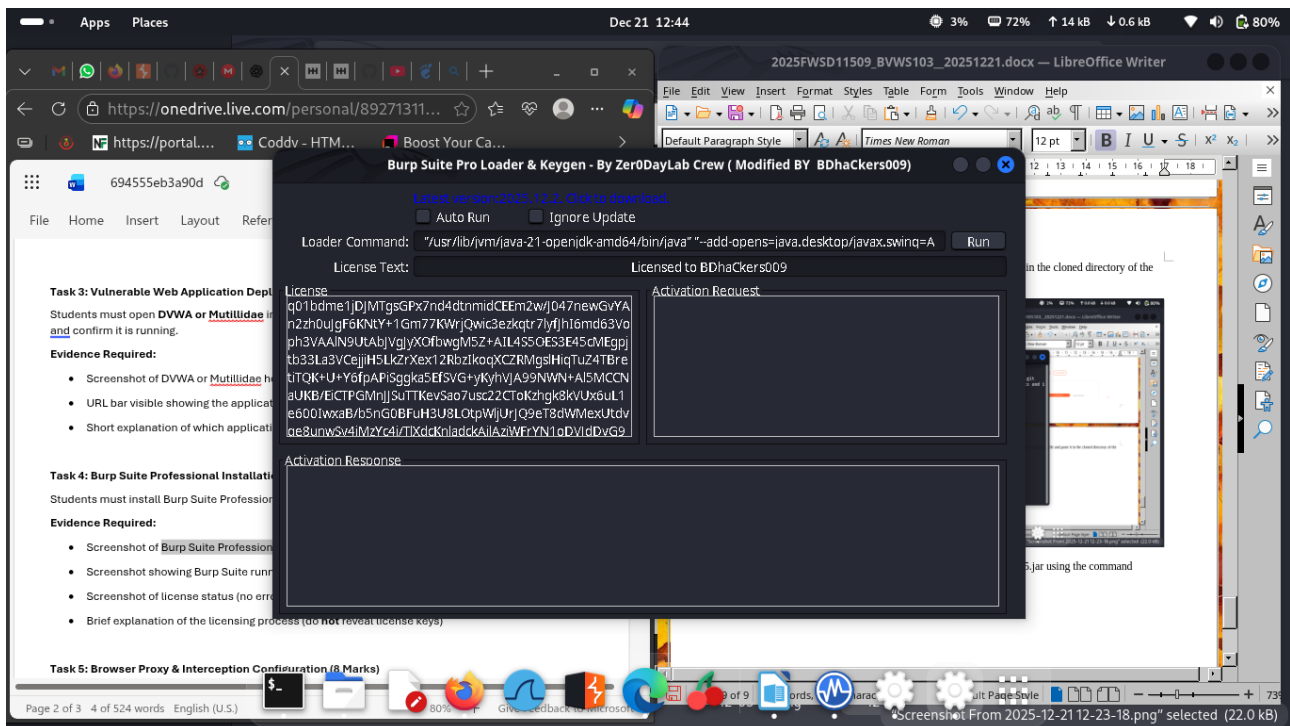
After finishing downloading the file I copy the file and paste it in the cloned directory of the burp suit



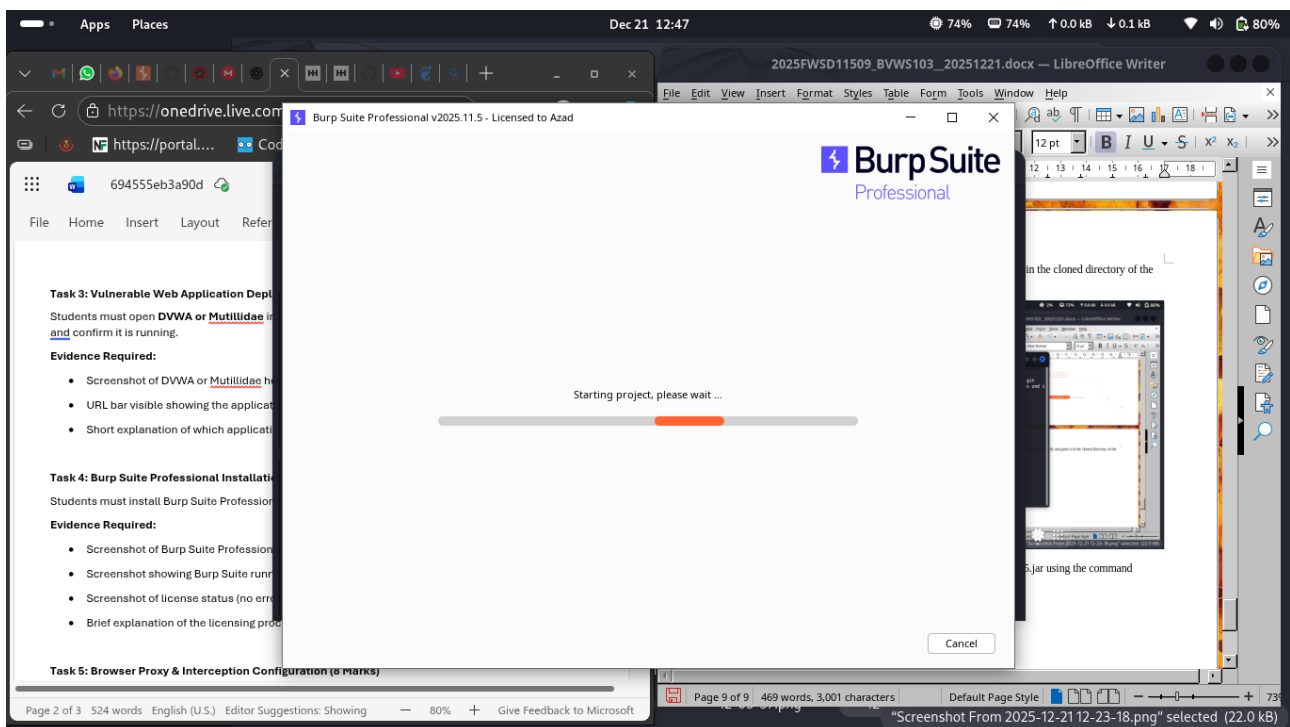
Then you use the burploadner.jar to open the burpsuite_pro_v2025.11.5.jar using the command `java -jar burploadner.jar`



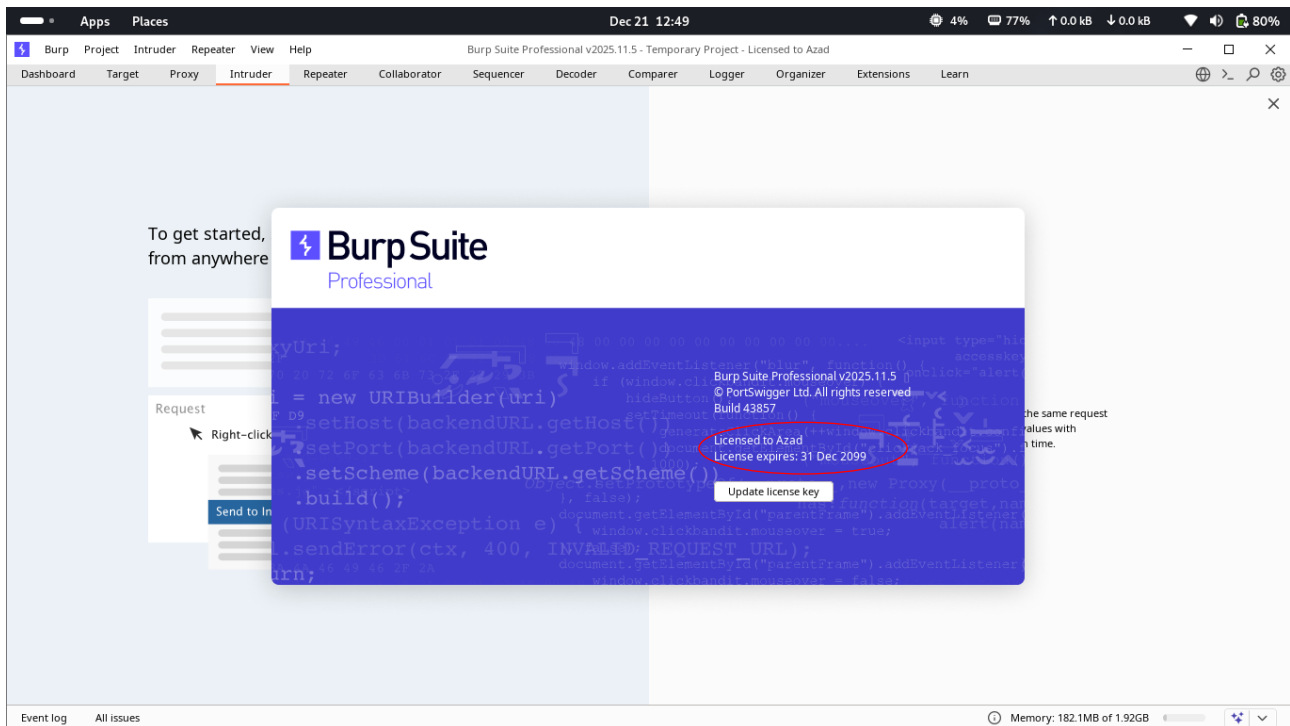
The burploader setting up the license



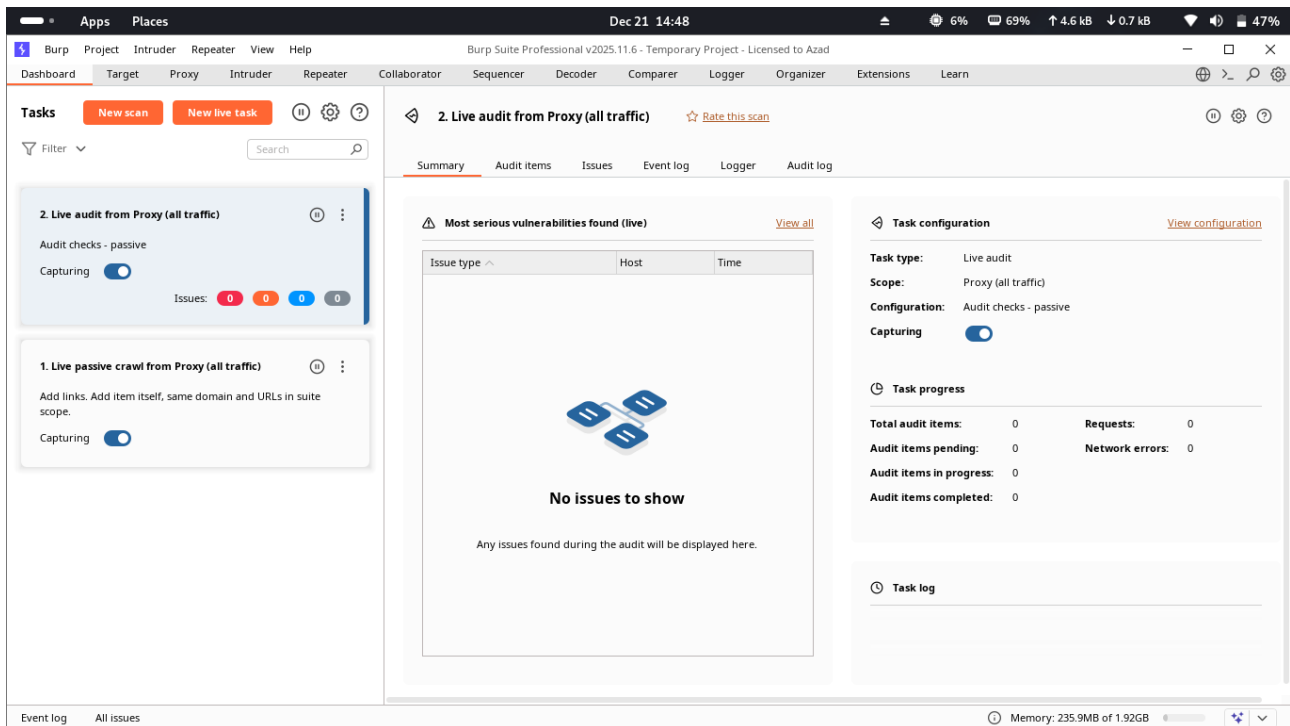
After all process of licensing the BurpSuite is stating perfectly



The license status showing no errors or expiration warnings

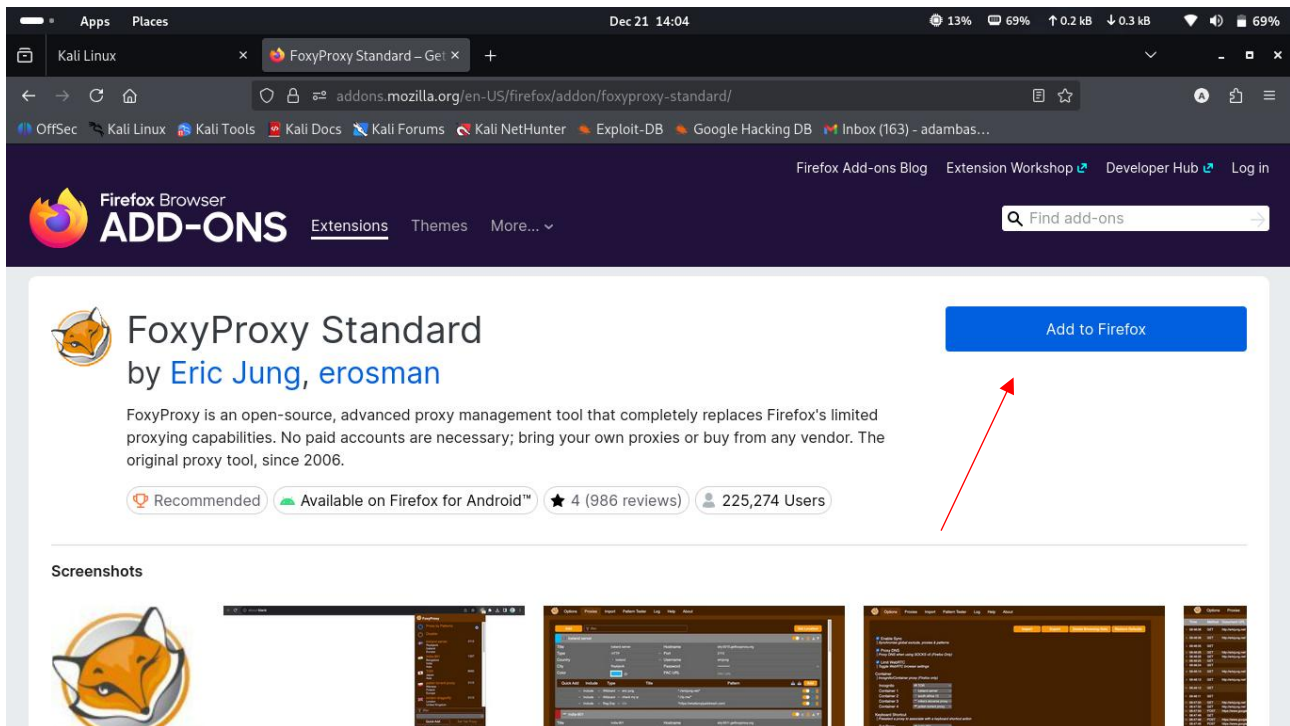


My burpSUIT darshbord

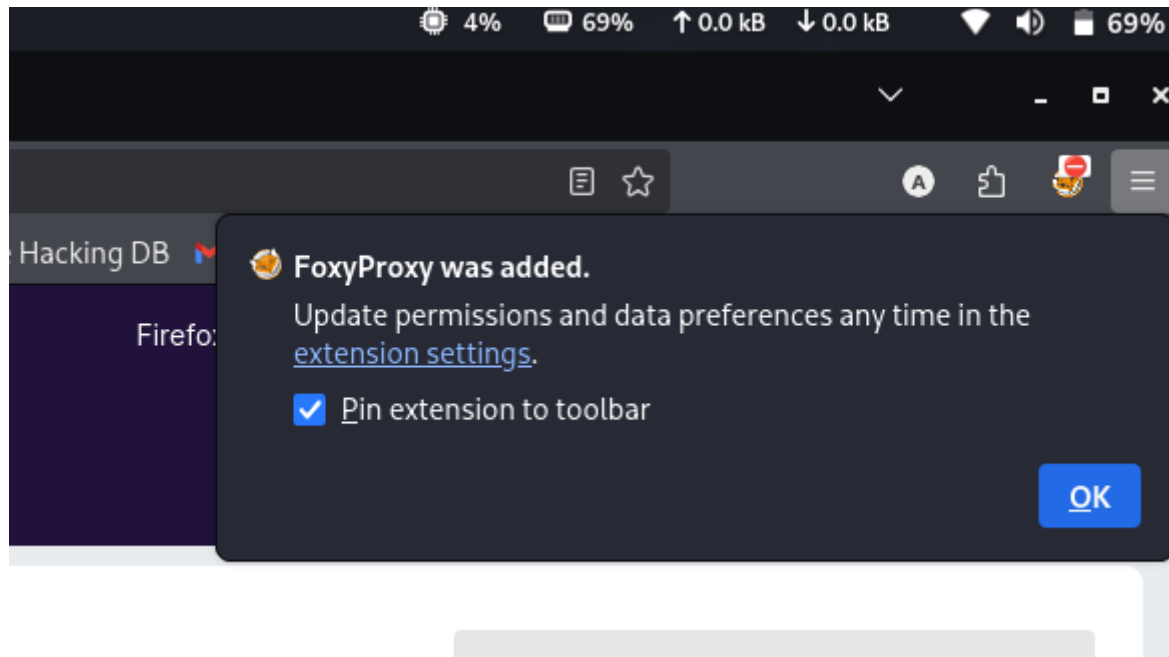


Task 5: Browser Proxy & Interception Configuration

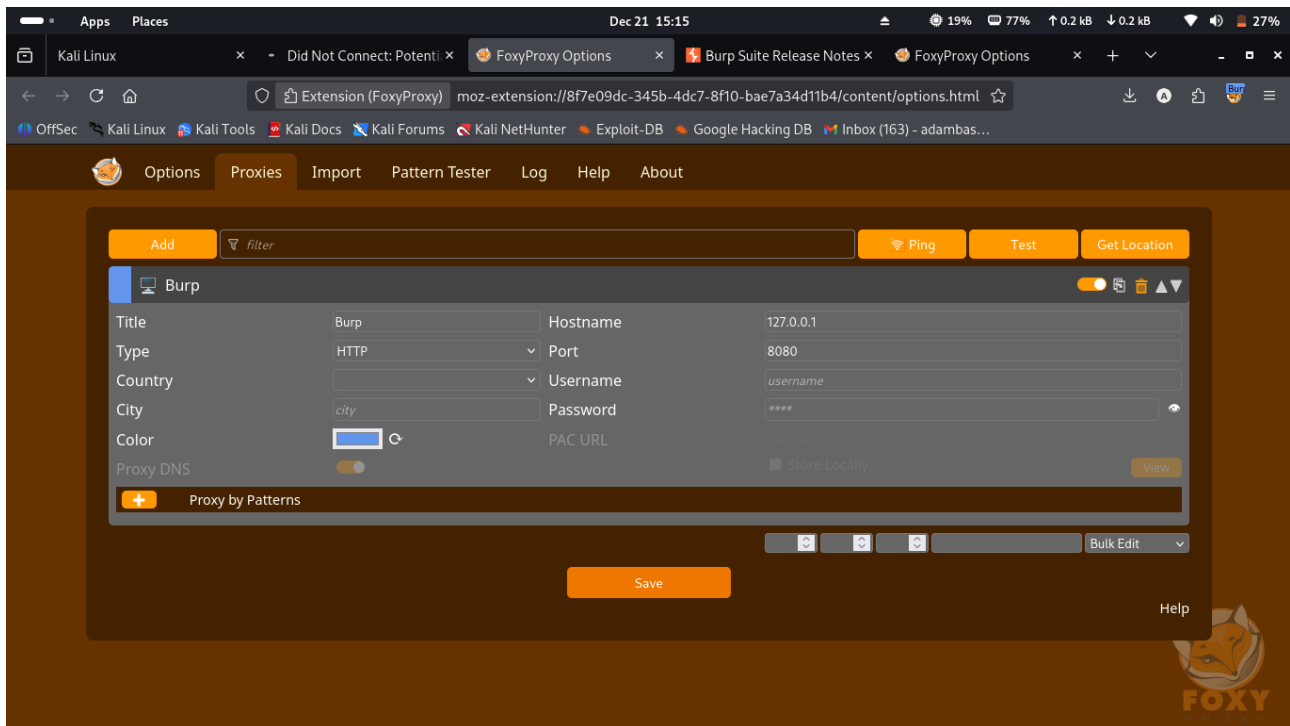
From this side we add the extension of our FoxyProxy to the Browser



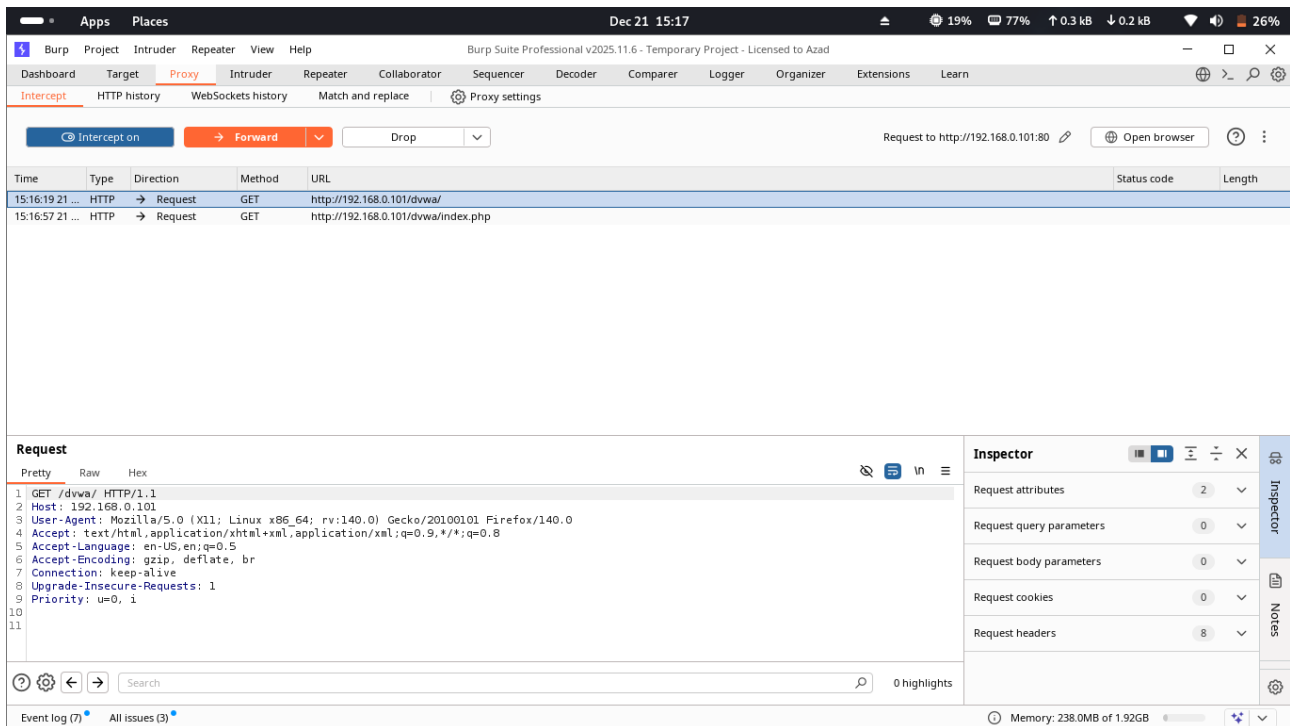
FoxyProxy was added successfully



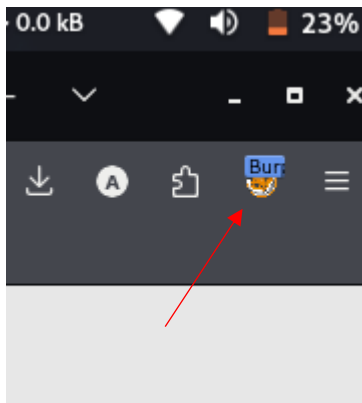
Setting up my local host connectivity using the ip 127.0.0.1 and port 8080



example of request send from the browser to the BurpSuite



The icon below shows that burp is on and ready to accept request



Conclusion

At the end of this lab Assignment I was able to:

- Build a functional penetration testing lab
- Verify attacker-target connectivity
- Deploy vulnerable web applications
- Configure professional web proxy tools
- Intercept and analyze web traffic